BoardGameGeek Data Analysis

What makes a great game and what's the impact of recognition?

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Background

BoardGameGeek (BGG) is the world's largest board game database and community platform, hosting over 180,000 titles rated by millions of enthusiasts worldwide. Each game record contains rich metadata including community ratings, complexity scores, player counts, playtimes, mechanics, categories, and ownership statistics. This dataset serves as the backbone of our analysis, providing the quality and popularity signals we use to evaluate game success.

This project brings together four data sources — BGG metadata, BGG resale marketplace listings, Wikipedia award records, and Google Trends data.

Motivation

We are a team of geeks interested in finding out which attributes impact how well received a board game might be. We are also curious to know which award publications prove most impactful to the success of products.

References

References are available on slide eleven.

Project Goals

1. **Which characteristics contribute to highly rated games?** Not all board games are created equal, and the BGG community rates them very differently. This question drives us to analyze game attributes and identify which design characteristics are most strongly associated with success.
2. **How can we define product success beyond BGG ratings?** A high average rating alone cannot fully capture a game's real-world impact, as it overlooks engagement, recognition, and market performance. Therefore, we constructed a composite success score combining BGG ratings, community comments, Game of the Year awards, resale data, and Google Trends data to measure product interest.
3. **Which characteristics contribute to a game's success?** Building on the composite success score defined in Q2, this question goes further by quantifying how much each game attribute — complexity, playtime, player count, and category — contributes to overall success through regression analysis. Rather than identifying correlations, the goal here is to measure the relative weight of each factor and determine which design decisions matter most.

Data Sources

BGG	Marketplace	Google Trend	Awards
Primary Dataset BGG games	**Secondary Dataset BGG resale listings**	**Secondary Dataset Google search metrics**	**Secondary Dataset Game of the year awards**
Size 44MB ~177K rows	**Size** 26MB ~ 328K rows	**Size** 29KB ~ 23 rows	**Size** 23KB ~ 466 rows
Important Variables complexity, num_ratings, min_players, max_players, min_playtime, max_playtime, min_age, avg_rating, num_comments, categories, num_owned, marketplace Extract	**Important Variables** id, name, date, price_orig, currency, condition, year_month	**Important Variables** Each column is a game's name (Total 389 columns)	**Important Variables** year, award, category, game_name
Access API	**Access** Extract from BGG dataset	**Access** API	**Access** Web Scraping
Snapshot 2026/05/27	**Snapshot** 2026/05/27	**Snapshot** 2026/05/27 (2004 - 2026)	**Snapshot** 2026/05/16
Link https://boardgamegeek.com/wiki/page/BGG_XML_API2	**Link** https://boardgamegeek.com/wiki/page/BGG_XML_API2	**Link** https://github.com/GeneralMills/pytrends	**Link** https://en.wikipedia.org/wiki/List_of_Game_of_the_Year_awards_(board_games)

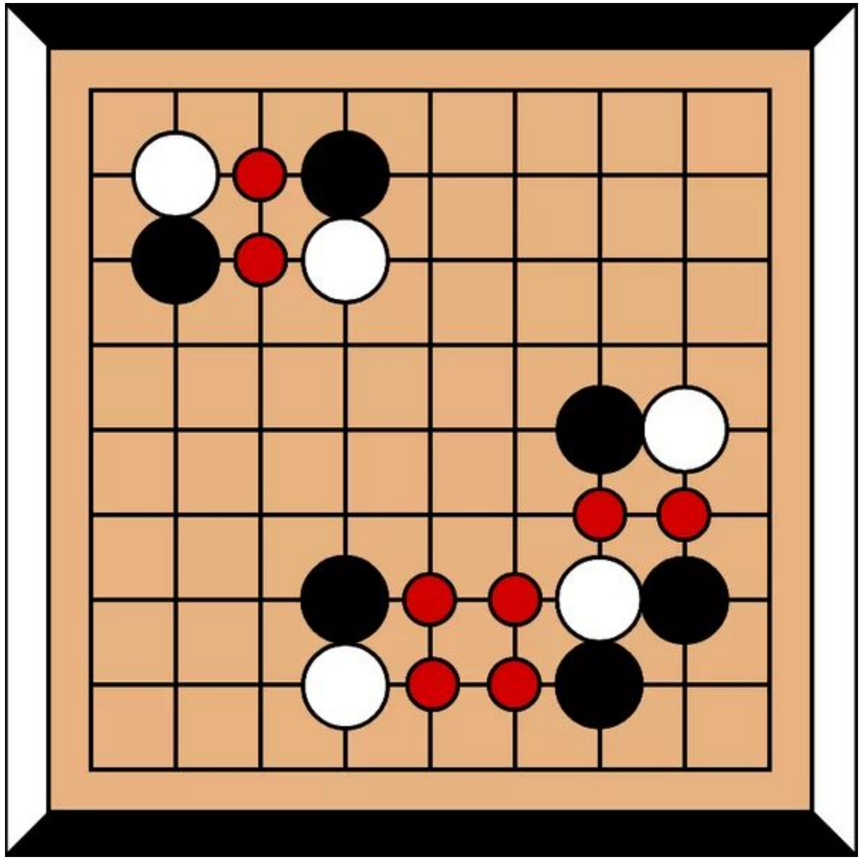
Data Manipulation

Cleaning

The Wikipedia tables for game of the year awards and Google trends data came in relatively cleanly, as they were smaller, more tightly controlled datasets.

For BGG game records we replaced 0 values for the **minimum players, maximum players, minimum playtime,** and **maximum playtime** attributes with a value of 1. Assumedly these are almost always mistakes, <u>most games can't be completed instantaneously or played autonomously</u>. If the minimum exceeded it's partner column in the case of players counts or playtime, it was replaced with the maximum to keep the data honest. Minefield was a great example of this, perplexingly listing 2,024 minimum players and just two maximum. Similarly we converted 0 values in three other columns with their averages: **years** was replaced with the median, **complexity** with the mean, and finally **bayes_rating** with the mean. **Minimum ages** greater than 21 were bound to a value of 21. Additionally we were forced to deal with some pretty extreme values in the maximum playtime and maximum players columns. In those cases we dropped records which listed 6,000 or more minutes to play and which allowed 1,000 or more players. Dropping records is always controversial, but it only lost us 73 games. Finally we dropped products with **categories** indicating an item was not a stand-alone game which amounted to a substantial chunk at 39,368 records. That's a sizeable bite out of the dataset, but we really aren't interested in interacting with supplementary content like expansions or add-ons.

For the BGG resale listings we only looked at a range of 1 dollar to 500 dollars after converting each listing to USD. Collector's items had extremely high values and component listings had very small ones; as the purpose of this exercise was to identify good products rather than monetarily valuable ones these ranges seemed appropriate. We only lost 3,745 listings outside of those upper and lower bounds.



Data Manipulation

Merging, Aggregation

Aggregating the Wikipedia data was straightforward when using the requests package. We pulled down the data from each table, mapped the **award name**, **year won**, and **game name** to their own columns from each table on the Wikipedia article.

Google Trends data carried its own difficulties. Google Trends data was collected for games identified as having won an award using the Pytrends package. The package works pretty well, but would bomb out if Google identified too many queries in a pattern or if the search term was found to carry very little or no search history. This required some babysitting to collect trend data from 2004-2026.

Similarly BGG data required a kludge of its own. BGG thankfully offers an API key requested via webform, however they limit calls on the “things” or products object to only 20 records at a time. To cache all 176,955 products required 10 hours of Chi-Hsun’s dutiful batching through all “thing” records in the database.

Fortunately the BGG “things” table and the BGG resale listings housed a matching numeric id for each individual product. That made for a very simple inner join. Joining any BGG data with Wikipedia and/or trends data was done by normalizing the product names, stripping white spaces, adjusting for case or special characters, and selecting for matches.

We cached all of our datasets to save reviewers the pain of re-aggregating our records as .csv files.

Analysis & Visualization

What Makes a Highly Rated Game?

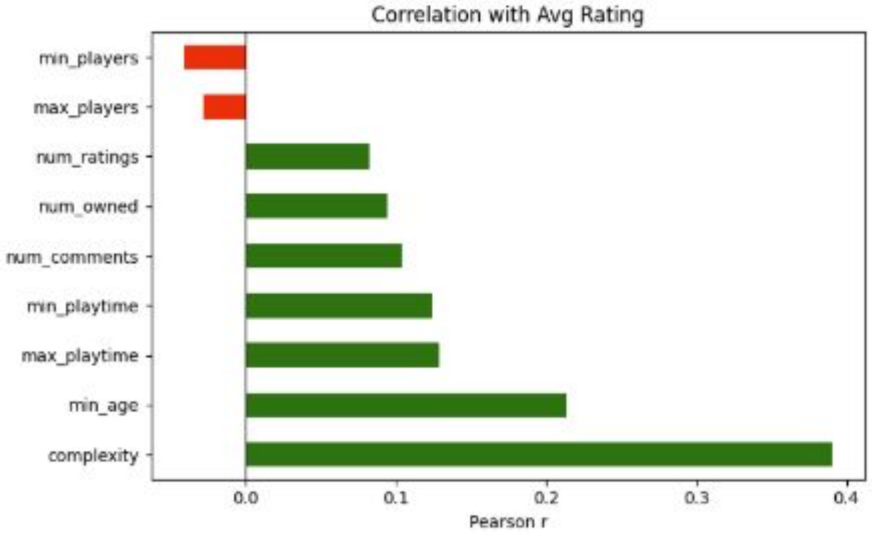


Figure 1. Correlation between Avg rating and other variables

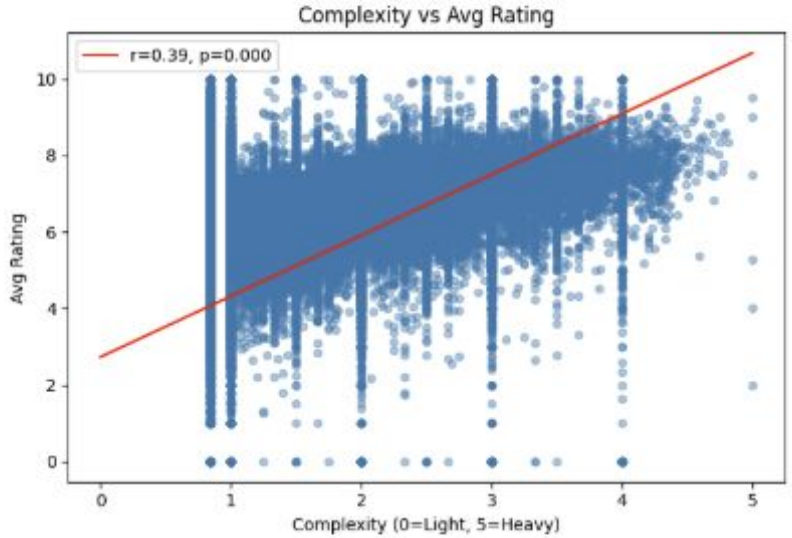


Figure 2. Complexity to Acg rating

This bar chart(Figure 1) examines which game attributes are most associated with higher average ratings on BoardGameGeek. To measure this, we use a correlation coefficient (**Pearson r**) — a number between -1 and 1 that tells us how strongly two variables move together, where a higher value means a stronger relationship. Among all features analyzed, **complexity** stands out as the strongest positive predictor (r = **0.39**), making complexity the **strongest** driver of a game's rating among all other features. Following **complexity**, **minimum age** and **playtime** also show modest positive associations. **Min_players** and **max_players** show slight negative correlation, suggesting that games requiring more players tend to receive lower ratings. Together, these findings suggest the BBG community favors **depth** over **accessibility**.

Since **complexity** showed the strongest relationship with rating, we take a closer look at it in the scatter plot(Figure 2). Each dot represents one game, plotted by its **complexity** score on the **x-axis** and its **average rating** on the **y-axis**, the red line shows the overall trend. The **p-value** of 0.000 confirms this relationship is statistically **significant** — not a result of random chance. The dots are widely spread — especially at **lower** complexity levels — meaning **light** games can still go either way, from beloved to poorly received. However, **heavier** games concentrate more consistently at higher ratings, reinforcing complexity as a reliable signal of quality within the BGG community.

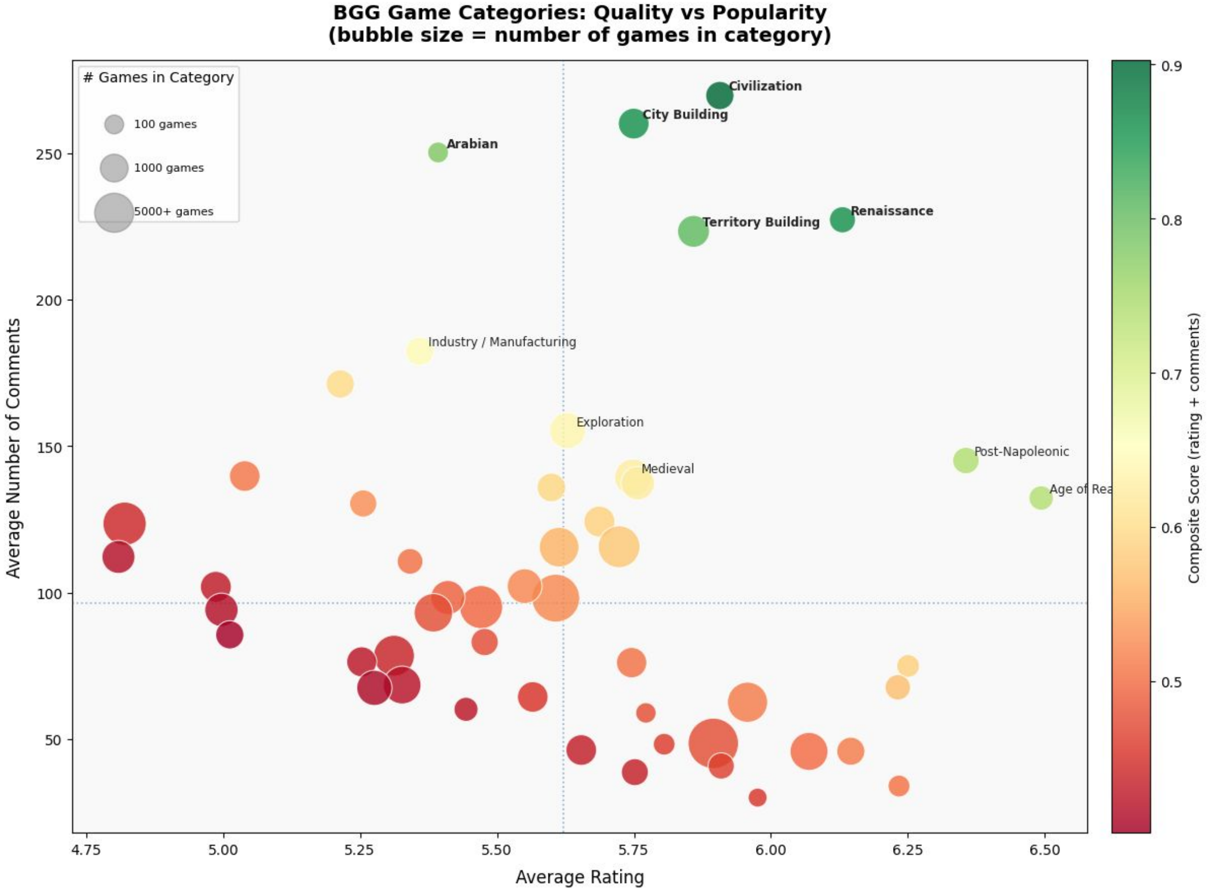
Analysis & Visualization

Which Game Categories Are Both Loved and Discussed?

This bubble chart plots the top 50 BGG game **categories** by their **average rating** (x-axis) and **average number of comments** per game (y-axis), where **comments** represent how engaged the community is with that category. Bubble size shows how many games belong to each category. **Color** reflects a composite score – both **average rating** and **average comments** are first scaled to a 0–1 range so they are comparable, then combined with equal weight (50% / 50%). Meaning both variables contribute equally to the composite score. The greener the bubble, the better the category performs on both dimensions combined. The dotted lines mark the median rating and median comment count, dividing the chart into four quadrants.

Civilization, City Building, Territory Building, and Renaissance stand out in the top-right region, having above-median ratings and the highest community engagement — averaging between 225 to 270 comments per game. These are the categories where games are not only well-received, but actively discussed.

Arabian is an outlier: its rating (~5.4) falls below the median, but it draws the third-highest comment count (~250), suggesting a small but unusually passionate community. Conversely, Post-Napoleonic and Age of Reason achieve the highest average ratings on the chart (~6.4–6.5) but attract comparatively fewer comments, indicating niche quality without broad engagement. An analysis of products themed around Other cultures might be interesting to investigate at a further date. The large red bubbles clustered near the center and bottom reflect high-volume categories.



Measuring Success

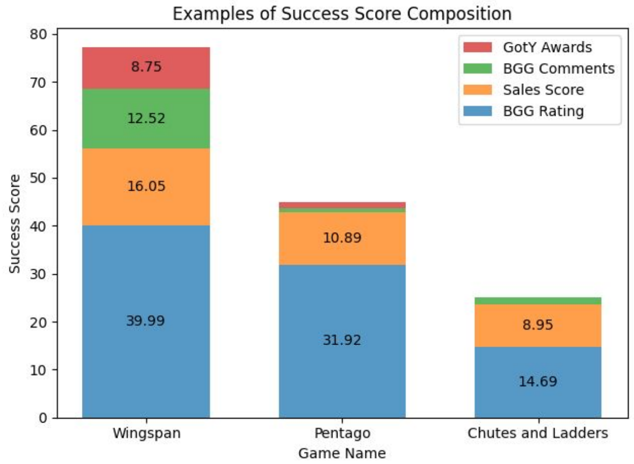
Augmenting BGG Ratings by Combining Multiple Metrics

Measuring Monetary Success

While the ratings compiled by BoardGameGeek provide insight into how well-received a game is, we wanted to expand on them by considering commercial success, social success, and industry recognition.

To measure commercial success, we calculated a sales score based on the secondhand marketplace data from BoardGameGeek. A simple sum of sales dollars would reward games that were released earlier, and a simple average might award newer, trendier games. In order to favor games with high average sales, but also reward games that have maintained those sales for multiple months, we calculated **sales score** using the formula below. We followed it with a min-max normalization:

$$\text{Sales Score} = \log\left(\text{Avg Monthly Sales}\right) \times \text{Months Active}^{\alpha}$$



Combining Indicators

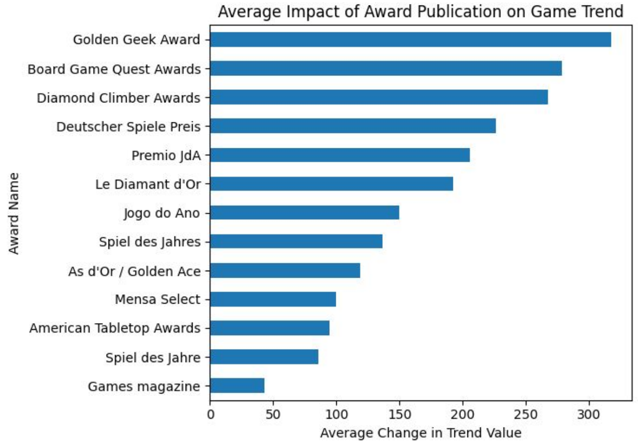
Additionally, we normalized a **count of comments** a game had on its BGG page and a **count of awards** a game had won as extra weight for our success score. The former represents how much social buzz a game has generated on BGG, and the latter is an indicator of recognition from major publications or game reviewers.

The normalized **BGG ratings** received the highest weight in calculating success score at 50%, then sales and comments weighted equally at 20%, followed by **Awards** at just 10% of the score.

Award Impact on Google Trends Data

Outside of these metrics, we investigated incorporating Trends data into the **success score**, but ultimately found it too arduous to collect for the breadth of BGG's dataset; specifics on that challenge can be found on slide four.

Unsurprisingly, BGG's own annual "Golden Geek Award" provided the greatest surge in average product hype according to the Google Trends data. This insight cements BGG.com's status as a hegemonic force in the online hobbyist board gaming community, at least within English-speaking countries that rely on Google most; Additionally it reinforced our decision to privilege BGG.com data when demarcating success.



Regression Analysis

Determining each Factors Effect on Game Success

Regression Data Subset

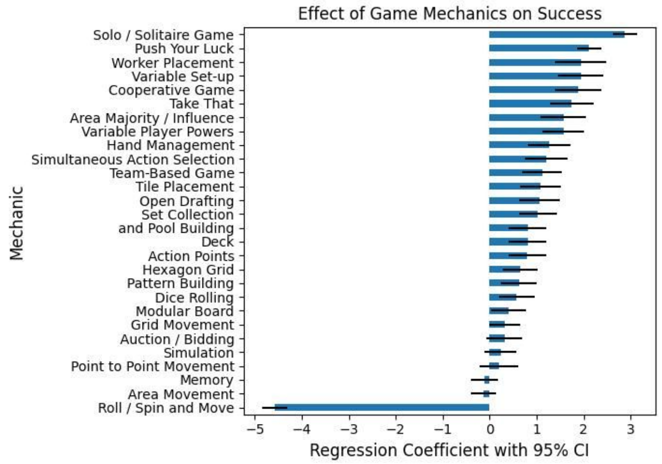
To enhance interpretability and limit the compute demands of the model generation, only the 28 most representative mechanics and categories were selected. The resulting training dataset contained 24,554 unique games, and the testing dataset contained 6,139 unique games.

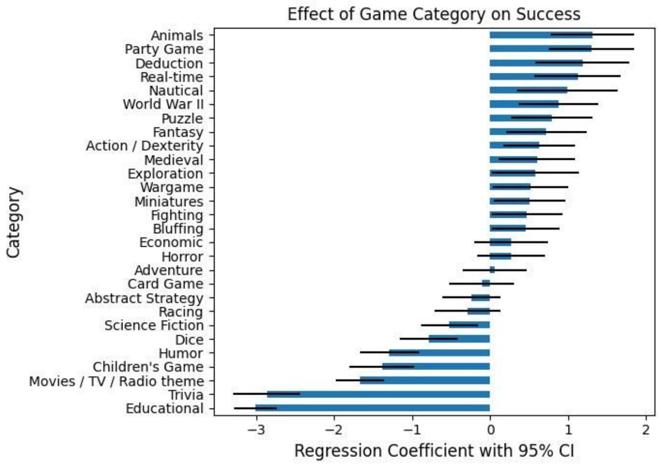
Training

Three different categories of variables were analyzed to determine why a game achieves success:

1. The basic **constraints** of each game - i.e. number of players, play time, age, and complexity
2. The different **mechanics** of games - i.e. solo or team-based, variable player powers, auctioning/bidding
3. Broad **categories** that a game could fall into - i.e. party games, fighting games, card games

Game constraints are continuous variables, meaning they could take on different values within a range, and model coefficients for these variables represent the predicted increase in success score per unit increase in variable value. Alternatively, game mechanics and categories are binary - the mechanic is either present in the game or it is not, and a game either falls into a category or it does not – in making predictions there is either no effect from these variables, or the effect is equal to the regression coefficient.





Performance

The model achieved an **R-squared of 0.419**, meaning that less than half of the variance in success between games can be explained by the variables included in our model.

The model has a Mean Absolute Error **(MAE) of 0.0138**, meaning that the model missed by an average of 0.0138 success points when predicting scores on the sample dataset reserved for testing.

Insights

Mechanics predictors with the largest absolute coefficients (Solo, Push Your Luck, Roll / Spin and Move) have narrow confidence intervals, whereas confidence intervals for category predictors are wider. <u>In general, mechanics impact success more reliably than categories.</u>

The "**Roll / Spin and Move**" mechanic has the greatest impact on success, albeit a negative impact, among all variables. This mechanic is used by game geeks derogatorily to refer to games that offer players little to no agency.

Of the game constraints variables (not charted), only **complexity** and **minimum players** had a significant impact on success. <u>A one point increase in complexity (a 0-5 scale) increased success by approximately 4 points, and each additional minimum player leads to a decrease in success of about 0.5 points.</u>

Limitations & Ethics

What was Potentially Missed and What could the Impact Be?

Other than the socio-economic and ethical limitations detailed on the right, we were forced to drop some records as described on slide three. In addition, some products from the Awards table simply did not generate enough hype in the Western dominated internet sphere to track on Google Trends; eight award winning products specifically.

In the context of our pricing data, many games represented in the BGG resale market were over represented by a select few. 27,187, or a little more than half of the games selected for analysis, had 2 or fewer listings. Even more products out of the 176,955 we pulled using our API did not have any listings at all.

Additionally, the data compiled for our analyses has glaring blind spots on more subjective aspects of game design. Traits such as **art style**, **writing**, and **manufacturing quality** certainly contribute to product success too. Regrettably these qualities were not measured in our selected datasources, but would prove interesting for future analysis.

Ultimately, we do not believe these limitations were impactful enough to discredit our findings. Many of the datasets and attributes proved bias in some fashion, but our decision to use weighted metrics sourced from multiple datasets proved effective. We believe that we designed a successful model which can accurately delineate qualities of product popularity in the online hobbyist Board Game community.

As a team we recognize that BGG data over-represents affluent, Western ideas and predilections regarding entertainment. The online hobbyist community does not equate to all people who enjoy board games all of the time. This is evidenced by the time and financial cost needed to engage in online discourse or to participate in maintaining the database itself; over-sampling these communities can amplify already over privileged opinions on gaming even further. Google Trends and WIkipedia data carry similar biases in participation as well. Fortunately, we are still of the belief that our analysis warrants a very low impact on marginalized communities in comparison with other data-oriented projects.

Efforts were made to not publish user specific data, but we do recognize that privacy is imperilled even in aggregation. Not every BGG user would consent to our findings and our calculations effectively surveil the BGG collective; in example, we made an inference on the BGG community's Geographic tastes on slide six when discussing the Arabian outlier, but this is not necessarily representative of all users. Even with these ethical dilemmas in mind, we still believe that the risk of harm is low enough to warrant publishing our results.

Conclusion

What Can we Infer about Product Success using These Data Sources?

What Makes a Game Successful?

Among all game attributes analyzed, **complexity** stands out as the dominant factor — the more a game demands from its players, the better it tends to be rated. This pattern holds consistently across the dataset, pointing to a BGG community that rewards **challenge** and **strategic depth** over **accessibility**. Rather than broad appeal, it is the willingness to invest time and effort that appears to define a well-regarded game — one that leaves a lasting impression on its players.

Board games can achieve success with a variety of mechanics and categories. Being single-player or having a single-player mode is highly correlated with success, along with the more strategic mechanics.

The most reliable way to make a game worse is by leaning heavily into luck-based mechanics. Less “serious” game categories, like trivia, children’s games, humorous games, and those based on existing IP from movies or TV tend be less successful than original works in categories with lots of action or a unique setting.

Which Game Categories are Loved and Discussed? What Conclusions can we Draw?

Not every game category performs the same way. Some attract both critical appreciation and active discussion, while others generate volume without engagement. **Strategy-themed** categories such as **Civilization** or **City Building** consistently occupy the high-performing end, while categories with the highest number of titles do not necessarily translate volume into quality or community engagement. While causality cannot be confirmed from the chart alone, the pattern is consistent — certain themes appear to cultivate both quality and community in ways that others do not.

Hypothetically, all of the attributes above and to the left line up with what we anticipated the BGG community would represent; people interested in collecting board games and participating in online discussions about them. <u>There is an interest in historical tropes, bias for challenge, desire for a product that allows engagement if alone, and love of games which do not punish players at random.</u> We invite our readers to parse the top and bottom ranked products on BoardGameGeek.com sorted by their proprietary Geek Rating. We suspect our metrics of success will not map perfectly, but that our findings will prove similar.

Statement of Work

Task	Daniel Buccos	Chi-Hsun Fan	Samuel Meisel
Data Sourcing	Support	Responsible	Support
Data Cleaning	Responsible	Support	Support
Data Merging	Responsible	Support	Support
Data Manipulation / Feature Engineering	Support	Responsible	Support
EDA	Support	Support	Responsible
Statistical Analysis / Modeling	Support	Responsible	Support
Data Visualization	Support	Support	Responsible
Data Validation / Quality Checks	Responsible	Support	Support
Documentation & Reproducibility	Responsible	Support	Support
Project Lead	Responsible	Responsible	Responsible

Referenced Works

+ I. Piotrowiak. (2025, March 17). number_of_voters_estimation, Version 6.
  - https://www.kaggle.com/code/jastrzt2/number-of-voters-estimation
  - referenced for data cleaning: "The 'Min time' and 'Max time' features exhibit an enormous range with a relatively small number of records for higher values. Based on the shapes of their distributions, we concluded that using the standard deviation method would be inappropriate and would still retain many potentially outlying data points."
+ Samarasinghe, Dilini & Barlow, Michael & Lakshika, Erandi & Lynar, Timothy & Moustafa, Nour & Townsend, Thomas & Turnbull, Benjamin. (2021). A Data Driven Review of Board Game Design and Interactions of Their Mechanics. IEEE Access. PP. 1-1. 10.1109/ACCESS.2021.3103198.
  - https://www.researchgate.net/publication/353787592_A_Data_Driven_Review_of_Board_Game_Design_and_Interactions_of_Their_Mechanics
  - This team conducted extensive descriptive analysis on game mechanics and constraints from BGG data. Their analysis of mechanics co-occurrence in games was especially useful in making decisions regarding variable inclusion for our regression analysis..